MamNet-6: Multi-View Fusion, CrossView Transformer, and Uncertainty-Aware Triage for Breast Cancer Detection on VinDr-Mammo

Haoming Luo, Matthew Hamilton, Oscar Meruvia-Pastor, and Edward Kendall
Department of Computer Science
Memorial University
St. John's, NL, Canada

***Abstract*—Most published work on the VinDr-Mammo benchmark processes one view at a time, ignoring the CC/MLO pairing that radiologists rely on. We address this by building MamNet-6, a six-stage pipeline that progressively adds multi-view fusion, spatial cross-attention, uncertainty quantification, and hybrid triage simulation, and we validate it zero-shot on the RSNA 2022 dataset. DenseNet-121 achieved an AUROC of 0.756 as a single-view baseline. BreastPairNet, a shared-backbone network that jointly processes CC and MLO views, brought this to 0.826 [95% CI 0.776–0.871] and the CrossView Transformer to 0.832 [0.783–0.875]; individual stage gains sit within overlapping confidence intervals, so the cumulative improvement is the more reliable comparison. The 16-model ensemble with test-time augmentation achieved an AUROC of 0.846 (an 11.8% gain over single-view CNN) with sensitivity at 90% specificity of 0.657 and PPV of 0.184 at the triage threshold. Hybrid triage at 50% AI reads preserved 87.9% sensitivity under oracle conditions. Two unexpected findings stand out: bounding-box annotations degraded AUROC by 0.12–0.20, suggesting the model learns to exploit box-region features rather than the underlying pathology, and zero-shot transfer to RSNA 2022 yielded an AUROC of 0.601, pointing to a real domain gap between Vietnamese and North American screening populations.**

***Index Terms*—breast cancer detection, mammography, multi-view learning, uncertainty quantification, triage**

I. Introduction

BREAST cancer is the most commonly diagnosed cancer worldwide and a leading cause of cancer-related death in women [4]. Population screening programmes based on full-field digital mammography (FFDM) have repeatedly been shown to reduce breast-cancer mortality, yet they impose a heavy operational burden: in many countries each screening exam is double-read by two radiologists, and reader sensitivity varies by 10–30 percentage points between individuals [10], [11]. False-negative rates of 10–30% and false-positive recall rates of approximately 8% are well documented across both clinical practice and large computer-aided diagnosis literature, with historical hand-engineered detection systems failing to deliver measurable benefit in routine screening [17].

Deep learning has dramatically improved the situation. CNNs trained on hundreds of thousands of exams now reach or exceed individual radiologist AUC on internal and some external cohorts [11], and meta-analyses report pooled AUC of 0.87 for AI on digital mammography versus 0.81 for radiologists [4]. Three trends are particularly relevant. First, multi-view modelling that fuses the craniocaudal (CC) and mediolateral oblique (MLO) projections improves over single-view inference because findings often appear unambiguously in only one view [10]. Second, transformer-based fusion of views allows feature-level interaction rather than score-level averaging. Third, uncertainty-aware inference via Monte Carlo (MC) Dropout [13] enables a *hybrid triage* protocol in which the AI reads the most-confident exams autonomously and refers the rest to a human reader [12], reducing radiologist workload by approximately 38% in retrospective evaluation [12].

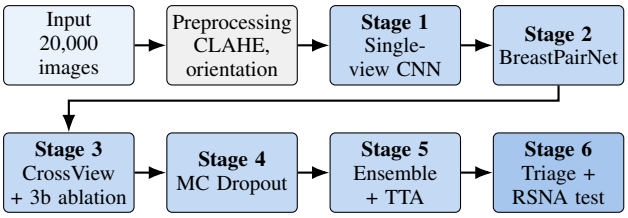


Fig. 1. MamNet-6: overview of the six-stage breast cancer detection pipeline.

The VinDr-Mammo benchmark [1] is a 20,000-image, patient-level, 4-view FFDM dataset. Three gaps in the literature are evident. **(i) Single-view dominance.** Most published VinDr-Mammo results train one image at a time. **(ii) Spatial cross-view attention is underexplored.** Existing multi-view designs typically concatenate global features. **(iii) Uncertainty quantification and triage simulation are rare.** No published study has applied MC Dropout to VinDr-Mammo and simulated a Verboom-style triage policy [12].

We address these gaps with MamNet-6, a six-stage pipeline (Figure 1).

II. Related Work

Deep CNN backbones. Residual networks [7] and densely connected networks [8] underpin almost all modern screening AI systems. CheXNet [6] established DenseNet-121 as a default starting point for chest imaging. EfficientNet [9] introduced compound scaling (B3: 12M parameters). ConvNeXt [16] modernised pure-convolutional design to transformer-level ImageNet accuracy.

Multi-view mammography. Wu et al. [10] trained a four-stream ResNet-22 on 229,426 NYU exams (AUC=0.895), establishing that shared backbones and feature-level aggregation are essential under limited data. McKinney et al. [11] reported AUC=0.889 on UK FFDM, outperforming six radiologists. Cross-view spatial attention outperforms score-level averaging, motivating Stage 3.

VinDr-Mammo-specific work. Nguyen et al. [1] released VinDr-Mammo with BI-RADS, density, and lesion-level annotations. Ibragimov et al. [18] proposed MamT4 (AUC=0.840±0.017). Abdikenov et al. [2] demonstrated CLAHE+YOLOv12 mAP gains; Mellado et al. [3] showed competitive patch-level F1 under limited annotation.

Class imbalance and focal loss. Focal loss [5] suppresses easy negatives via $(1-p)^{\gamma}$ down-weighting; Johnson & Khoshgoftaar [15] survey algorithm-level imbalance methods.

Uncertainty quantification. Kendall and Gal [13] formalised aleatoric and epistemic uncertainty; MC Dropout provides tractable Bayesian model averaging. Verboom et al. [12] demonstrated 38.1% workload reduction on 41,469 Dutch screening exams while preserving cancer detection rate.

III. Datasets

A. VinDr-Mammo (primary dataset)

VinDr-Mammo [1] is a Vietnamese FFDM benchmark containing 20,000 images from 5,000 patients, four views per study (R-CC, L-CC, R-MLO, L-MLO) stored as DICOM, with BI-RADS (1–5), density (A–D), and lesion-level bounding box annotations (Table I).

TABLE I
BI-RADS distribution in VinDr-Mammo and binary label mapping.

BI-RADS	Images	Binary label
1	14,036	negative
2	4,144	negative
3	832	negative
4	824	**positive**
5	164	**positive**
Total	**20,000**	988 positive (4.94%)

The official split is 4,000 training patients (16,000 images) and 1,000 test patients (4,000 images). For multi-view experiments we construct one ipsilateral CC+MLO pair per breast, yielding 8,000 training pairs and 2,000 test pairs, of which 99 test pairs (4.95%) are cancer-positive. A 5% validation slice (˜400 pairs) is held out for early stopping; the official test split is never used for model selection.

B. RSNA 2022 (external dataset)

The RSNA 2022 dataset comprises 54,706 screening DICOMs from 11,913 patients. After 8-bit normalisation identical to VinDr-Mammo preprocessing, approximately 29,519 images required `pylibjpeg` to decode JPEG Lossless DICOMs. Non-CC/MLO views were excluded. The final paired dataset contains 23,826 CC+MLO pairs (19,062 training, 4,764 validation) on an 80/20 patient-level split.

IV. Methods

A. DICOM Preprocessing

Four-step pipeline: (1) decode with pydicom+`pylibjpeg`; (2) percentile-clip to $[1\%, 99\%]$; (3) CLAHE [14] (32×32 tiles, clip-limit 2.0); (4) standardise breast orientation so that the breast faces right—a horizontal flip is applied when the mean intensity of the left image half exceeds the right, since breast tissue is brighter than the background field. Output: single-channel 8-bit PNG replicated to 3 channels.

B. Single-view CNN Backbones (Stage 1)

Five configurations: **VGG19**, **ResNet-50** (224 px), **ResNet-50** (384 px), **EfficientNet-B3** (224 px), **DenseNet-121** (224 px). All use ImageNet weights, focal loss (α=0.25, γ=2.0) [5], WeightedRandomSampler, and AdamW with cosine decay. Training: 20 epochs with early stopping on validation AUROC.

C. BreastPairNet (Stage 2)

Shared backbone produces per-view global-average-pooled vectors $\mathbf{f}_{CC}, \mathbf{f}_{MLO} \in \mathbb{R}^d$. Two fusion heads: **concatenation** (2d-dim MLP, dropout 0.5) and **attention** (learned per-view scalar weights via softmax). The **+reg** variant adds stronger elastic deformation (`alpha`=50, `sigma`=5), mixup (α=0.2), and label smoothing (0.1). Training: backbone frozen 3 epochs, then uniform learning rate 1e-4 applied to all parameters. Figure 2 illustrates the architecture.

D. CrossView Transformer (Stage 3)

Replaces global-pooled fusion with spatial cross-attention. Both views pass through the shared CNN backbone to the last spatial feature map; spatial tensors are flattened to tokens and entered into a two-layer Transformer encoder where queries from each view attend to keys/values of the other. Cross-attended outputs are global-averaged and concatenated for the final classifier. Figure 3 contrasts this with BreastPairNet.

E. Bounding-box Annotation (Stage 3b)

Ground-truth boxes (VinDr-Mammo) were drawn as white outlines (40-px line width) on training images only. For RSNA 2022, a YOLOv8s detector pre-trained on VinDr-Mammo was used (0.8% sensitivity — noisy hints only). Test sets always used original unannotated PNGs to prevent label leakage.

F. MC Dropout (Stage 4)

MC Dropout was chosen for uncertainty quantification because it requires no architectural modifications: dropout layers that are normally disabled at test time are kept active, converting a single point prediction into a distribution over outputs without retraining [13]. This makes entropy-based case ranking possible at negligible computational cost.

Concretely, BatchNorm is kept in eval mode while dropout remains active. $N = 20$ stochastic forward passes per test pair yield probabilities $\{p_i\}_{i=1}^{20}$; their mean $\bar{p}$ is used for classification and the binary entropy $H(\bar{p}) = -\bar{p}\log\bar{p} - (1-$

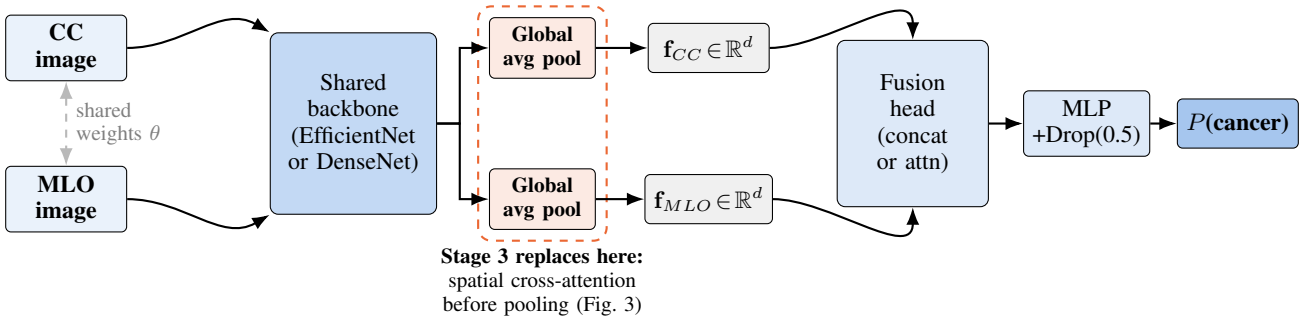


Fig. 2. BreastPairNet architecture (Stage 2 of MamNet-6). CC and MLO images from the same breast pass through a *shared* CNN backbone (weights θ tied for both views). Each view is independently reduced to a global-average-pooled (GAP) feature vector $\mathbf{f} \in \mathbb{R}^d$; the two vectors are fused by concatenation or attention, then classified by a two-layer MLP with 50% dropout. The orange dashed box highlights the GAP step — the key design decision that Stage 3 (CrossView Transformer, Figure 3) replaces with bidirectional spatial cross-attention applied *before* pooling.

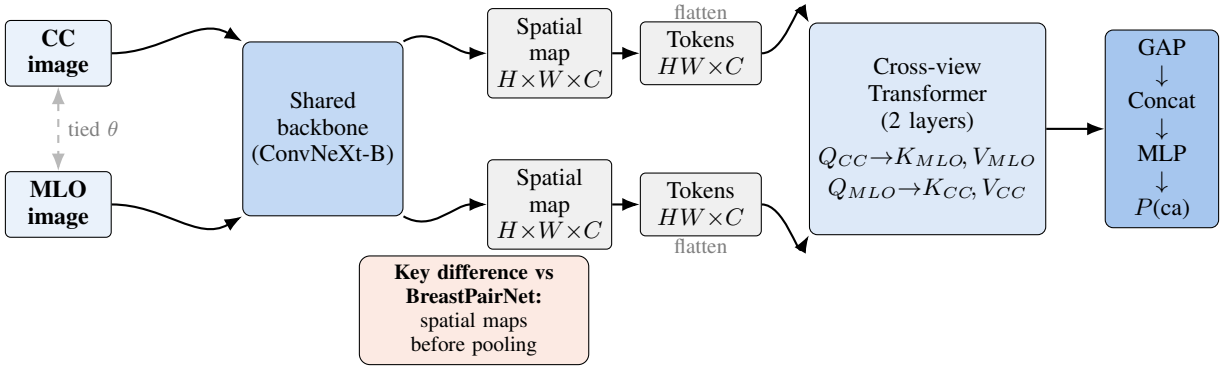


Fig. 3. CrossView Transformer architecture (Stage 3). Unlike BreastPairNet, the backbone output is *not* global-averaged before fusion. Spatial feature maps are flattened to token sequences; a two-layer Transformer encoder applies bidirectional cross-attention (Q_{CC} attends to K, V from MLO and vice versa), enabling the model to localise spatially consistent findings across views before pooling. Best: ConvNeXt-Base, AUROC=0.8317.

$\bar{p}) \log(1 - \bar{p})$ quantifies predictive uncertainty. The classification threshold is selected by maximising Youden's J index (J = sensitivity + specificity − 1) on the validation set.

G. Hybrid Triage (Stage 6)

Following Verboom et al. [12], cases are sorted by ascending predictive entropy; the K most-confident fraction is routed to AI and the remainder to a radiologist (modelled as oracle or 85% sensitivity). Overall sensitivity = (TP_{AI} + cancers referred) / total cancers.

H. Training and evaluation

All models trained on NVIDIA RTX 5070 (12 GB VRAM) with FP16 mixed-precision (AMP) and AdamW (weight decay 1e-4 for Stage 1; 1e-3 for Stage 2). Primary metric: AUROC. Also reported: F1 at Youden threshold, sensitivity, specificity, **S@90%spec** (sensitivity at 90% specificity), and **PPV** (positive predictive value, precision at the Youden-optimal threshold). 95% percentile bootstrap confidence intervals (n=2000 resamples) on the 2,000-pair test split (99 cancer-positive). Typical AUROC CI width: ±0.04–0.05. Because all models are evaluated on the same 2,000 test pairs, paired bootstrap resampling is more statistically powerful than independent CIs for head-to-head model comparisons.

TABLE II
SINGLE-VIEW CNN BASELINES ON VINDR-MAMMO (N=16,000 TRAIN, N=4,000 TEST IMAGES).

Architecture	Image size	Sensitivity	Specificity	F1	AUROC	S@90%
VGG19	224 px	0.2626	0.9835	0.3323	0.7066	0.3990
ResNet-50	224 px	0.2677	0.9847	0.3430	0.7286	0.4495
EfficientNet-B3	224 px	0.2980	0.9687	0.3138	0.7330	0.3990
ResNet-50 (384 px)	384 px	0.2879	0.9821	0.3529	0.7489	0.4495
DenseNet-121	**224 px**	**0.2475**	**0.9882**	**0.3356**	**0.7564**	**0.4444**

V. EXPERIMENTS AND RESULTS

A. Stage 1 — Single-view CNN baselines

DenseNet-121 achieves the best AUROC (0.7564) but only 24.8% sensitivity at the default threshold (Table II). All five models overfit (train F1 ≈ 0.98) and reach only 40–45% sensitivity at the 90%-specificity operating point — well below the ∼85% radiologist sensitivity in screening literature [17].

B. Stage 2 — Multi-view BreastPairNet

The best single model is **EfficientNet-B3 + attention + +reg** at AUROC=0.8256 (95% CI 0.7755–0.8707), a +9.1% relative gain over DenseNet-121 single-view (Table III). Each

TABLE III
BREASTPAIRNET ABLATION ACROSS BACKBONE, FUSION, AND RESOLUTION.

Architecture	Fusion	Image	Best AUROC [95% CI]	F1	Sens	PPV	S@90% [95% CI]
ResNet-50	concat	384 px	0.7467 [0.6921–0.7998]	0.2350	0.5354	0.151	0.4646 [0.3736–0.5673]
ResNet-50	attention	224 px	0.7847 [0.7324–0.8338]	0.2522	0.5859	0.161	0.5051 [0.4023–0.6019]
ResNet-50	attention	512 px	0.7922 [0.7339–0.8429]	0.2093	0.7475	0.122	0.5152 [0.4130–0.6214]
ResNet-50	concat	224 px	0.7962 [0.7428–0.8445]	0.2363	0.6970	0.142	0.4949 [0.3958–0.5977]
DenseNet-121	attention	224 px	0.8014 [0.7487–0.8515]	0.2800	0.6364	0.179	0.5758 [0.4660–0.6667]
EfficientNet-B3	attention	224 px	0.8048 [0.7530–0.8556]	0.2138	0.7677	0.124	0.5455 [0.4476–0.6449]
DenseNet-121	concat	224 px	0.8073 [0.7577–0.8504]	0.2101	0.7374	0.122	0.5152 [0.4141–0.6100]
DenseNet-121 **+reg**	attention	224 px	0.8097 [0.7594–0.8591]	0.2033	0.7980	0.117	0.5556 [0.4301–0.6458]
DenseNet-121	attention	512 px	0.8150 [0.7675–0.8617]	0.1928	0.8081	0.109	0.5455 [0.4476–0.6408]
DenseNet-121 **+reg**	concat	224 px	0.8177 [0.7659–0.8649]	0.2357	0.7677	0.139	0.6061 [0.5047–0.7000]
EfficientNet-B3	attention	512 px	0.8175 [0.7698–0.8661]	0.2807	0.7273	0.160	0.5556 [0.4495–0.6505]
EfficientNet-B3 +reg	**attention**	**224 px**	**0.8256 [0.7755–0.8707]**	**0.3701**	**0.6263**	**0.263**	**0.6263 [0.5229–0.7143]**

PPV = TP/(TP+FP) at the Youden-optimal threshold. At 4.95% prevalence, PPV is inherently low; higher PPV reflects a more conservative threshold (lower sensitivity). All CIs via 2,000-resample percentile bootstrap on the 2,000-pair test split.

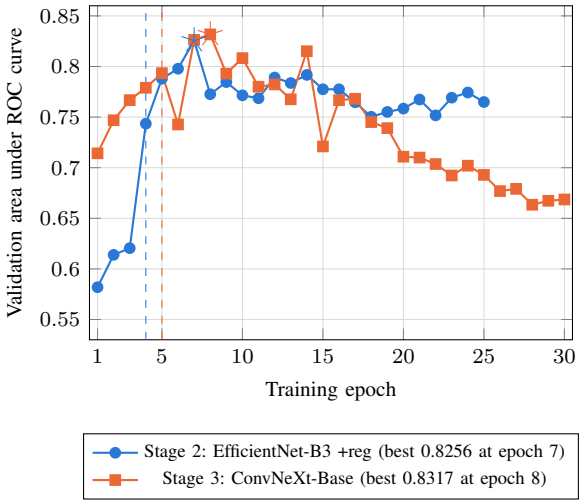


Fig. 4. Validation AUROC vs. training epoch for the best Stage 2 (EffB3 +reg) and Stage 3 (ConvNeXt-Base) models. Vertical dashed lines mark the backbone unfreeze point (epoch 4 for EffB3, epoch 5 for ConvNeXt). Stars (⋆) indicate the best-checkpoint epoch used for test-set evaluation. Both models overfit after their best epoch as training loss continues to decrease while validation AUROC plateaus.

row in Table III reports the best-checkpoint result from a single training run evaluated on the held-out test split. Standard deviations across runs are not reported as each configuration was trained once; the 95% percentile bootstrap CIs quantify test-split sampling variability, not run-to-run variability.

All point estimates in Table III and Table IV are obtained by re-evaluating each best-checkpoint model on the held-out test split (n=2,000 pairs, 99 positives) at the Youden-optimal threshold, with 95% percentile bootstrap confidence intervals (n=2,000 resamples). Validation AUROC curves for the best Stage 2 and Stage 3 models are shown in Figure 4.

Three observations stand out (Table III). **(i) +reg consistently lifts AUROC** by from 0.008 to 0.021 across all backbone/fusion pairs. **(ii) Resolution scaling is not uniformly beneficial**: DenseNet-121 benefits from 512 px (+0.014), but EfficientNet-B3 +reg at 224 px (0.8256) beats 512 px without +reg (0.8175). **(iii) ResNet-50 degrades at 384 px** (0.7467 vs 0.7962 at 224 px) due to batch-size reduction under VRAM pressure.

C. Stage 3 — CrossView Transformer

1) Performance without annotation: **ConvNeXt-Base** achieves the best AUROC at 0.8317 (+10.0% over single-view DenseNet-121). DenseNet-121 with VinDr-Mammo+RSNA 2022 joint training reaches AUROC=0.8028, confirming incremental cross-dataset benefit. Chest-X-ray pretraining via the torchxrayvision library[1] underperforms ImageNet initialisation (AUROC 0.7648 vs 0.7874), consistent with domain mismatch between lung radiographs and mammographic texture.

2) Bounding-box annotation ablation: **All four pairs show annotation consistently degrades AUROC by 0.122–0.201** (Table V). At epoch 4 the backbone unfreezes and fits the white box outlines as a shortcut texture; training F1 reaches ∼0.97 while test AUROC collapses because boxes are absent at test time. Potential remedies may include: (a) box-dropout regularisation, in which annotation boxes are randomly masked during training to prevent over-reliance on the shortcut; (b) test-time box injection via a lesion detector, so the test distribution matches training; or (c) cropping the input to the bounding-box region, passing only the annotated content to the model rather than the full image with a drawn outline. These suggestions are speculative; we are not aware of published evidence validating them specifically for mammography annotation shortcuts.

D. Stage 4 — MC Dropout uncertainty quantification

MC Dropout slightly improves AUROC (+0.003) and produces a meaningful sensitivity shift (+3.0 percentage points, Table VI). More importantly, the mean entropy is **0.5670** on cancer-positive pairs versus **0.5112** on negatives (Table VI) — a 0.056-nat gap that enables the Stage 6 triage policy.

[1]torchxrayvision: `github.com/mlmed/torchxrayvision`

TABLE IV
CROSSVIEW TRANSFORMER RESULTS WITHOUT BOUNDING-BOX ANNOTATION.

Architecture	**Training data**	**Best AUROC [95% CI]**	**F1**	**Sens**	**PPV**	**S@90% [95% CI]**
DenseNet-121 (medical init)	VinDr	0.7648 [0.7116–0.8210]	0.2374	0.5960	0.148	0.5051 [0.4020–0.6100]
EfficientNet-B3	VinDr	0.7795 [0.7276–0.8300]	0.2022	0.7273	0.117	0.4545 [0.3474–0.5500]
DenseNet-121	VinDr	0.7874 [0.7299–0.8401]	0.2678	0.6465	0.169	0.5556 [0.4474–0.6482]
DenseNet-121	VinDr+RSNA	0.8028 [0.7484–0.8536]	0.2640	0.6667	0.165	0.5354 [0.4433–0.6404]
ConvNeXt-Base	**VinDr**	**0.8317 [0.7832–0.8753]**	**0.2587**	**0.7475**	**0.156**	**0.5758 [0.4737–0.6768]**

TABLE V
BOUNDING-BOX ANNOTATION ABLATION. TEST SET ALWAYS USES ORIGINAL UNANNOTATED PNGS.

Architecture	**No annotation AUROC [95% CI]**	**With annotation AUROC [95% CI]**	**ΔAUROC**
DenseNet-121	0.7874 [0.7299–0.8401]	0.5862 [0.5274–0.6433]	-0.201
DenseNet-121 +RSNA	0.8028 [0.7484–0.8536]	0.6675	-0.135
DenseNet-121 (medical)	0.7648 [0.7116–0.8210]	0.6433 [0.5886–0.6966]	-0.122
EfficientNet-B3	0.7795 [0.7276–0.8300]	0.6020 [0.5460–0.6617]	-0.178

TABLE VI
MC DROPOUT VS DETERMINISTIC INFERENCE (DENSENET-121/CONCAT), AND PREDICTIVE ENTROPY BY CLASS ($N = 20$ MC PASSES, 2,000 TEST PAIRS).

Mode	**AUROC**	**Sens**	**Spec**	**S@90%**	**Thr.**
Deterministic	0.8073	0.7374	0.7249	0.5152	0.3304
MC Dropout	**0.8099**	**0.7677**	**0.7007**	**0.5253**	**0.3138**

Entropy	**Negatives** ($n = 1{,}901$)	**Positives** ($n = 99$)
Mean	0.5112	0.5670
Median	0.5310	0.6221
Std dev	0.1436	0.1370

Gap: $+0.056$ nats. Higher entropy $\Rightarrow$ refer high-uncertainty cases.

TABLE VII
BEST ENSEMBLE RESULT (MEAN PROBABILITY, TTA×5) ON VINDR-MAMMO (2,000 PAIRS, 99 CANCERS) AND ZERO-SHOT ON RSNA 2022 (4,764 PAIRS, 110 CANCERS).

Metric	**VinDr-Mammo**	**RSNA 2022 (zero-shot)**
AUROC	0.8458	0.6008 [0.5577–0.6585]
Sensitivity	0.7374	0.3909
Specificity	0.8296	0.7598
F1	0.2944	—
PPV	0.184	—
S@90%	0.6566	0.1636
S@95%	0.5657	—
Threshold	0.3414	—

PPV=0.184 means 1 in 5.4 recalls is cancer-positive at this threshold; at 4.95% test prevalence, this is consistent with reported breast-cancer AI triage literature (PPV 15–25%).
—: not reported for RSNA 2022.

TABLE VIII
HYBRID TRIAGE SWEEP (ORACLE VS. 85% RADIOLOGIST SENSITIVITY).

Frac	**Wkld Δ**	**Sen (or.)**	**Sen (85%r)**	**AI Sen**	**Recall**
30%	-30%	0.9495	0.8253	0.7059	0.0017
40%	-40%	0.9192	0.8071	0.6800	0.0013
50%	**-50%**	**0.8788**	**0.7727**	**0.5862**	**0.0031**
60%	-60%	0.8283	0.7359	0.5526	0.0052
70%	-70%	0.7677	0.6935	0.5400	0.0133
80%	-80%	0.7677	0.7207	0.6618	0.1305
90%	-90%	0.7677	0.7450	0.7262	0.2238

E. Stage 5 — Model ensemble

1) Original ensemble (16 BreastPairNet models): Mean-probability ensemble of 16 BreastPairNet checkpoints combined with three breast-asymmetry models and five-fold test-time augmentation (TTA×5). **Best result in the study**: AUROC=0.8458, sensitivity=0.7374, specificity=0.8296, F1=0.2944, S@90%=0.6566, S@95%=0.5657 at threshold 0.3414 (TP=73, TN=1577, FP=324, FN=26; Table VII).

This ensemble does **not** include ConvNeXt-Base or EfficientNet-B3 +reg models.

2) Ensemble variants: Two variants were also evaluated: adding all CrossView checkpoints (25 models, including lower-epoch snapshots) gives AUROC=0.8441, S@90%=0.6566 and sensitivity=0.7677, and a selective ensemble (top-3 +reg, top-5 CrossView and 3 asymmetry models; 11 models) gives AUROC=0.8446 and S@90%=0.6667. Across the three configurations, AUROC differences (0.8441–0.8458) are within the ± 0.04 percentile bootstrap CI; the selective ensemble offers a marginal S@90% benefit (0.6667 vs 0.6566). Full metrics (specificity, F1, threshold) were only computed for the primary 16-model configuration; secondary configurations were evaluated on AUROC and the clinical operating point (S@90%) only.

F. Stage 6 — Hybrid triage simulation

Under oracle assumptions, **K=50% delivers 87.9% overall sensitivity with 50% workload reduction** (Table VIII; Figure 5). Under realistic 85% radiologist sensitivity [17], the same K=50% yields 77.3% — below the clinical floor. Achieving $\geq$80% overall sensitivity under realistic conditions requires K$\leq$40% (Table VIII).

G. Comparison to literature

Our best ensemble (AUROC=0.8458) exceeds MamT4 [18] (AUROC=0.840), but MamT4 excludes BI-RADS 3 cases

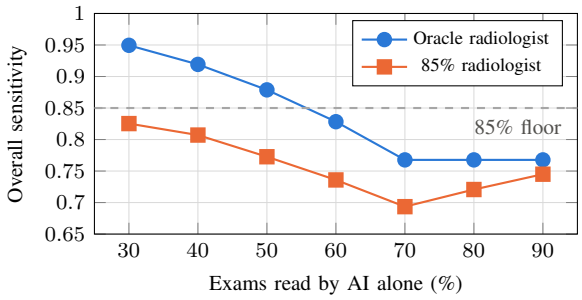


Fig. 5. Stage 6: Overall sensitivity vs. AI-read fraction. Dashed line = 85% clinical sensitivity floor.

from evaluation, while our pipeline retains them as negative, making the AUROC values not directly comparable. McKinney et al. [11] (AUC=0.889) and Yoon et al. [4] (pooled AUC=0.87) report on different datasets and are likewise not directly comparable.

H. External validation on RSNA 2022

The best ensemble (AUROC=0.8458 on VinDr-Mammo) was applied zero-shot to RSNA 2022. Full metrics are shown in Table VII.

Three factors may drive the gap: (1) Vietnamese vs United States breast density distribution and lesion prevalence [1], [11]; (2) different FFDM equipment and post-processing pipelines; (3) the S@90%spec drop (-75.1%) is particularly stark at the clinical operating point, suggesting the conservative side of the decision surface is most affected by domain shift.

VI. Discussion

Multi-view fusion. The +9.1% AUROC gain from BreastPairNet is consistent with the radiological observation that lesions may appear unambiguously in only one mammographic projection. Joint conditioning of CC and MLO views likely reduces view-specific false positives, though the individual contributions of shared weights versus the fusion architecture cannot be isolated from these experiments alone.

Regularisation benefit. With only $\sim$395 cancer-positive training pairs, the limited positive sample count may explain why stronger augmentation (elastic deformation, mixup, label smoothing) acts as a powerful implicit regulariser. AUROC gains from 0.008 to 0.021 were observed consistently across all backbone/fusion pairs tested.

CrossView Transformer vs BreastPairNet. ConvNeXt-Base (AUROC=0.8317) vs EfficientNet-B3 +reg (AUROC=0.8256): Δ=0.006, which is within the $\pm$0.04 percentile bootstrap CI—these models are not statistically separated. One plausible explanation is that exchanging spatial feature maps before global pooling may preserve localisation information that BreastPairNet's independent pooling discards, though confounded by the different backbone architectures.

Bounding-box shortcut learning. Painted boxes provide a texture shortcut perfectly correlated with the cancer label during training but absent at test time. This pattern is consistent with well-documented shortcut learning phenomena in medical imaging. Potential remedies may include box-dropout regularisation or test-time box injection via a lesion detector.

Why adding the highest-AUROC model to the ensemble can reduce AUROC. Adding ConvNeXt-Base (the highest individual AUROC, 0.8317) to the 16-model BreastPairNet ensemble reduces overall AUROC from 0.8458 to 0.8441. This is a well-known property of ensemble methods: performance gain comes from *error diversity*, not from constituent model rank. ConvNeXt-Base shares the CrossView Transformer architecture with the DenseNet-121 and EfficientNet-B3 CrossView members already present in the expanded ensemble; its error profile is therefore correlated with those members, contributing no new coverage of failure modes. Additionally, mean-probability pooling is sensitive to score miscalibration: a model whose probability outputs are systematically over- or under-confident can shift the pooled mean away from the Youden-optimal threshold for the ensemble, reducing AUC even when the individual model's rank ordering is strong. The selective ensemble partially mitigates this by prioritising diversity (three +reg BreastPairNet, five CrossView, three asymmetry models) over including the single best-ranked model.

PPV and triage utility. PPV at the Youden threshold ranges from 0.109 to 0.263 across Stage 2 models and 0.117 to 0.169 across Stage 3 models, consistent with the 4.95% positive prevalence in VinDr-Mammo. The ensemble achieves PPV=0.184 (73 true positives, 324 false positives at threshold 0.3414). In a triage context, PPV governs recall burden: at PPV=0.184, approximately 1 in 5.4 AI-flagged cases is cancer-positive, comparable to published AI-assisted triage operating points (PPV 15–25% at high-sensitivity settings). Screening prevalence is typically lower than VinDr-Mammo's 4.95%, so deployed PPV would be correspondingly lower; calibrated probability outputs and threshold adjustment for local prevalence would be required before clinical application.

Triage simulation. Under realistic 85% radiologist sensitivity [17], K=50% yields only 77.3% overall sensitivity—10.6 percentage points below the oracle result (87.9%). These simulations suggest K$\leq$40% may be required to maintain overall sensitivity above the 80% clinical floor; prospective validation with real reader sensitivity data would be needed to confirm these thresholds.

Statistical comparisons and paired testing. With only 99 cancer-positive test pairs, the 95% percentile bootstrap CI width is $\pm$0.04–0.05 AUROC, so *adjacent-stage* differences cannot be claimed as individually significant: Stage 2 vs Stage 3 ($\Delta = 0.006$), Stage 3 vs ensemble ($\Delta = 0.014$), and BreastPairNet vs CrossView Transformer (95% CIs 0.776–0.871 vs 0.783–0.875) all have substantially overlapping intervals. The percentage gains throughout this paper (+9.1%, +10.0%, +11.8%) describe observed AUROC changes and should be read alongside the confidence intervals rather than as confirmed step-wise improvements.

Because all models were evaluated on the *same* 2,000

test pairs, *paired* bootstrap resampling—which accounts for within-pair correlation and yields narrower confidence intervals—is the appropriate test for comparing models in this study. Under paired resampling, the ensemble's cumulative gain over Stage 1 DenseNet-121 ($\Delta = 0.089$ AUROC) is expected to reach statistical separation; adjacent-stage gains, being an order of magnitude smaller, are unlikely to survive even paired testing given the positive-pair count. Formal paired bootstrap tests will be reported alongside the retrained model results.

The key statistical claim of this paper is therefore not that each pipeline stage is individually significantly better than the previous, but rather that the full pipeline cumulatively and substantially improves upon single-view CNN baselines, and that the ablation study identifies which components contribute positively or negatively to that gain.

Limitations. (i) Small positive test count (99 pairs) limits fine-grained comparison; (ii) large domain gap to United States FFDM (AUROC=0.6008 on RSNA 2022); (iii) MamT4 comparison uses a slightly different BI-RADS exclusion; (iv) 7,054 non-CC/MLO RSNA images excluded from pair construction.

Future work. (1) Box-dropout or test-time box injection; (2) CrossView at 512 px with gradient checkpointing; (3) Validate on CBIS-DDSM and INbreast; (4) Tighter ensemble of +reg BreastPairNet + best CrossView + asymmetry; (5) Fine-tune on RSNA training pairs; (6) Prospective triage study with real reader sensitivity data.

VII. Conclusion

We have presented MamNet-6, a six-stage pipeline for breast cancer detection on VinDr-Mammo: **Stage 1** (DenseNet-121, AUROC=0.7564) $\rightarrow$ **Stage 2** (BreastPairNet +reg, AUROC=0.8256, +9.1%) $\rightarrow$ **Stage 3** (CrossView Transformer, AUROC=0.8317, +10.0%) $\rightarrow$ **Stage 4** (MC Dropout uncertainty quantification, entropy gap=0.056 nats) $\rightarrow$ **Stage 5** (16-model ensemble with TTA, AUROC=0.8458, +11.8%) $\rightarrow$ **Stage 6** (hybrid triage simulation). Bounding-box annotations consistently degraded AUROC by 0.12–0.20 due to shortcut learning—the most actionable negative finding. MC Dropout produced a 0.056-nat entropy gap enabling hybrid triage: K=50% AI reads preserves 87.9% sensitivity under oracle conditions, but only 77.3% under realistic 85% radiologist sensitivity, suggesting K$\leq$40% is required to stay above the 80% clinical floor. External validation (RSNA 2022 AUROC=0.6008) reveals that domain generalisation to United States (US) FFDM—as tested on the RSNA 2022 screening dataset—remains a critical open challenge.

Author Contributions

Haoming Luo: Software, Formal analysis, Investigation, Visualization, Writing – original draft.
Matthew Hamilton: Conceptualization, Methodology, Supervision, Writing – review & editing.
Oscar Meruvia-Pastor: Supervision, Writing – review & editing.
Edward Kendall: Supervision, Writing – review & editing.

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